

# VELLANKI NITISH BALA SOURI RAJ

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## SUMMARY

Pre-final year B.Tech CSE student (CGPA: 9.11/10) with hands-on experience in building web applications, working with databases, writing automation scripts, and deploying software using cloud and containerization tools. Comfortable working with Python, JavaScript, and databases like PostgreSQL and MongoDB. Familiar with cloud concepts, server-side programming, and AI tools. Ranked 61st out of 4,000+ teams at NEC 2025 (IIT Bombay).

## EDUCATION

**Vignan's Institute of Information Technology**

Visakhapatnam, India

*B.Tech in Computer Science & Engineering – CGPA: 9.11 / 10*

*2023 – 2027*

## TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript, TypeScript, Java

**Web & Backend Development:** Node.js, Express.js, FastAPI, React.js, REST APIs

**Databases:** PostgreSQL, MongoDB, Redis, SQL (MySQL, SQLite)

**Cloud & Infrastructure:** Docker (containerization), CI/CD pipelines, Vercel, Render, Microsoft Azure (AZ-204)

**AI & Machine Learning:** scikit-learn, Deep Learning (Neural Networks, CNNs), OpenCV, Python scripting, LangChain, AI workflow automation

**Mathematics & Science:** Linear Algebra, Probability & Statistics, Calculus, Data Structures & Algorithms

**Tools:** Git, Postman, Drizzle ORM

**Certifications:** Microsoft Azure AZ-204 | Lyzr AI Agent Architect

## EXPERIENCE

**AI & Prompt Engineering Intern**

Jun 2025 – Jul 2025

*VaultofCodes*

*Remote*

- Designed 30+ templates for AI-based automation workflows, improving task accuracy by around 25%.
- Connected AI outputs to REST API backend pipelines to automate repetitive tasks.
- Reduced AI processing costs by about 20% by optimizing how inputs were structured.

**AI & ML Intern**

Apr 2025 – Jun 2025

*GDG – EduSkill AP*

*Visakhapatnam*

- Built an image classification system using Python and OpenCV, achieving around 92% accuracy.
- Improved model performance by 8% through better image preprocessing and feature extraction.

## PROJECTS

**Skill Connect – Web Platform** | *React.js, Node.js, PostgreSQL* | 

- Built a web platform with secure JWT login, database optimizations (40% faster queries), and automated CI/CD deployment.
- Ensured zero-downtime deployments using automated pipelines on cloud infrastructure.

**URL Shortener Service** | *Node.js, PostgreSQL, Redis, Docker* | 

- Built and deployed a backend service handling 1,000+ simultaneous requests with response times under 50ms.
- Reduced database load by 60% using a Redis caching layer; cut response time from 120ms to 45ms.
- Containerized the full application using Docker for consistent deployment across environments.

**SafeTrip AI – Travel Safety Platform** | *React, Node.js, PostgreSQL* | 

- Built a full-stack web application with user authentication, trip planning, and emergency response features.
- Designed 10+ REST API endpoints for authentication, trip management, and location-based services.
- Configured a relational database with 6 tables and automated setup scripts for development environments.

**Multi-Agent Trading Dashboard** | *Python, LangChain, Streamlit* | 

- Built a Python-based stock market analysis tool integrating live data from Binance and WazirX APIs.

- Created an interactive dashboard with real-time charts for market trend monitoring and decision support.

#### **Resume Screening Tool** | *Python, FastAPI, scikit-learn* |

- Built an AI-powered tool that automatically matches resumes to job descriptions, processing 50+ resumes per minute.
- Reduced manual resume review effort by around 70% using ML-based similarity scoring.

#### **Academic Monitor – Smart Attendance System** | *Python, OpenCV, BLE, OTP* | *In Progress*

- Developing a facial recognition attendance system using OpenCV, targeting less than 2% false-positive rate.
- Designing a multi-layer anti-proxy system using Bluetooth Low Energy (BLE) proximity checks and OTP verification.

### ACHIEVEMENTS & LEADERSHIP

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- Ranked 61 / 4,000+ teams in NEC 2025 (National Engineering Competition) conducted by IIT Bombay.
- CodeChef 2-Star rated – strong problem-solving skills in data structures and algorithms.
- Web Development Lead at EDCELL VIIT – managed web projects for the college entrepreneurship cell.